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Uncertainty-aware deep reinforcement learning for sustainable electric vehicle routing: A hybrid optimization framework

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ABSTRACT

Urban freight accounts for approximately 20% of transport-related greenhouse gas emissions, yet electric vehicle (EV) routing faces critical barriers: limited range, charging constraints, and energy uncertainty. Existing methods cannot quantify stranding risk or support chance-constrained battery feasibility, as energy consumption exhibits 30%–40% variability. We address the EVRPTW under operational uncertainty via probabilistic energy forecasting and hybrid optimization. Bidirectional LSTM networks deliver calibrated uncertainty quantification (94.7% empirical coverage for 95% confidence intervals), enabling model-based chance-constrained battery feasibility. Graph neural networks encode battery-aware spatial–temporal dependencies, while Proximal Policy Optimization with optional mixed-integer programming refinement balances real-time adaptability (0.3–0.8 s inference) with offline solution quality. Evaluation on 30 Solomon-derived EVRPTW benchmark instances (60–100 customers) shows an 8.8% cost reduction over the strongest learning baseline (MVMoE), rising to 12.2% under dynamic conditions. Real-world deployment with a 25-vehicle electric fleet over 13 weeks validates €143,000 in projected annual savings, 98.1% on-time delivery, 15.5% energy reduction, and 96% planning time reduction, with fleet emissions at approximate carbon parity with diesel on Italy's transitional grid alongside elimination of local air pollutants. Transfer learning protocols reduce region-adaptation retraining effort by ~95% (from ~72 h to 3–5 h of fine-tuning), requiring only 400–600 local routes. This work demonstrates a pathway toward more sustainable urban logistics through uncertainty-aware optimization with measurable environmental, economic, and operational benefits.

1. Introduction

The global transition toward sustainable urban systems has placed considerable focus on decarbonizing freight logistics, a sector responsible for approximately 8% of global CO_2 emissions and up to 20% of urban transport emissions [1,2]. Electric vehicle adoption in freight operations offers a compelling pathway toward decarbonization, with well-to-wheel emissions reductions of 40%–60% compared to conventional diesel vehicles [3,4]. Major logistics operators have announced ambitious electrification targets, and policy frameworks reinforce this transition, with the European Green Deal mandating zero-emission urban logistics in major cities by 2030 [5–7].

Despite this momentum, a critical operational barrier constrains EV deployment: **energy consumption uncertainty**. Real-world electric delivery operations exhibit 30%–40% consumption variability based on traffic conditions, weather parameters, payload variations, and terrain [8–10]. This uncertainty manifests in three operational risks that deterministic planning approaches cannot address: (1) vehicle stranding events when actual consumption exceeds predictions, compromising service reliability and requiring costly emergency interventions; (2) excessive conservatism when safety margins are set too high, reducing operational efficiency and negating economic advantages; and (3) inability to support model-based chance-constrained feasibility under calibrated uncertainty, limiting fleet operator confidence in electrification.

The Electric Vehicle Routing Problem with Time Windows (EVRPTW) extends classical vehicle routing by incorporating battery constraints, charging stations, and realistic energy consumption models [11,12]. Classical mathematical methods achieve high solution quality

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but become computationally prohibitive under uncertainty [13], while stochastic formulations require computational resources that preclude real-time decision-making [14]. Recent deep reinforcement learning approaches offer rapid inference [15,16] yet often lack explicit battery management and uncertainty quantification. While several approaches address subsets of these requirements [17–19], few frameworks simultaneously integrate probabilistic energy forecasting with formal uncertainty quantification, adaptive deep reinforcement learning, and hybrid optimization mechanisms that support both real-time replanning and offline solution refinement for electric vehicle routing under operational uncertainty.

Existing approaches fail to jointly address three critical requirements: (1) realistic modeling of energy consumption uncertainty with calibrated confidence intervals, (2) model-based chance-constrained feasibility, and (3) computational efficiency enabling both real-time replanning and high-quality offline optimization. This methodological gap constrains fleet operators to conservative strategies that sacrifice operational efficiency and fail to realize projected emissions reductions.

The central research question is: *Can uncertainty-aware deep reinforcement learning integrated with mathematical optimization enable electric vehicle routing that simultaneously achieves competitive solution quality, model-based chance-constrained feasibility, and computational tractability for operational deployment?* Addressing this requires advancing neural architectures that explicitly represent battery constraints, creating probabilistic energy models with calibrated confidence intervals, and designing hybrid protocols that balance learning and optimization.

This work advances uncertainty-aware electric vehicle routing through three primary contributions:

1. **Calibrated Probabilistic Energy Forecasting:** Bidirectional LSTM networks predict energy consumption distributions with uncertainty quantification achieving 94.7% empirical coverage for 95% confidence intervals, enabling model-based chance-constrained battery feasibility (conditional on the Gaussian arc-level energy model and on the calibration of the learned uncertainty estimator) rather than ad-hoc safety margins.
2. **Hybrid Uncertainty-Aware Optimization:** Integration of learned policies with mathematical refinement provides dual-mode operation — sub-second real-time replanning under uncertainty and high-quality offline optimization — through Graph Attention Networks for battery-aware encoding and Proximal Policy Optimization with optional MILP refinement.
3. **Deployment Validation and Transfer Learning:** Real-world validation demonstrates operational viability (13 weeks, 25-vehicle fleet) with quantified multi-dimensional sustainability impact, while transfer learning protocols reduce region-adaptation computational effort by approximately 95% (from ~72 h of full retraining to 3–5 h of fine-tuning; see Table 2), enabling practical adoption with only 400–600 local routes. This metric refers specifically to model-adaptation compute and data, and does not capture broader deployment effort components such as TMS integration, pilot validation, operator training, and operational acceptance, which are themselves discussed in Section 6. Full implementation is released at https://github.com/Aymen-Anbar/evrptw-repo to support reproducibility.

These contributions provide evidence of measurable environmental, economic, and operational benefits through both computational validation and real-world deployment, offering concrete guidance for fleet operators pursuing cost-effective electrification, cities working toward decarbonization objectives, and policy makers designing incentive structures—reconciling environmental mandates with operational realities.

2. Literature review

2.1. Electric vehicle routing: From deterministic to stochastic formulations

The Electric Vehicle Routing Problem with Time Windows (EVRPTW) has emerged as a critical extension of classical vehicle routing optimization, incorporating the unique operational constraints of battery-electric vehicles. Schneider et al. [11] established the foundational mathematical framework by introducing battery capacity constraints, recharging station visits, and nonlinear charging functions into the traditional VRP-TW formulation. Early exact solution approaches, including the branch-and-price framework by Desaulniers et al. [20], demonstrated optimality guarantees for small-scale instances (≤25 customers) but remained computationally intractable for realistic operational scales. Related work on charging infrastructure planning, such as the flow refueling location model (FRLM) framework by Park and Lee [13], addresses optimal charging station placement considering heterogeneous chargers and nonlinear charging times, though these location models assume fixed routing patterns rather than jointly optimizing routes and charging decisions.

Recognizing these computational limitations, researchers developed metaheuristic approaches that trade solution optimality for scalability. Keskin et al. [21] introduced partial recharging strategies that significantly reduced total travel time by allowing vehicles to recharge only as needed rather than to full capacity. Mavrovouniotis et al. [22] applied ant colony optimization with dynamic pheromone updating mechanisms tailored to battery state constraints. More recently, matheuristic approaches combining mathematical programming with heuristic search have achieved state-of-the-art performance: Bruglieri et al. [12] reported superior solution quality through Random Kernel Search that iteratively refines kernel solutions via mixed-integer programming, while Zhang et al. [23] employed column generation within a branch-and-price framework for instances with up to 100 customers.

However, a significant limitation in much of this classical optimization literature is the assumption of deterministic energy consumption. Real-world electric vehicle operations exhibit substantial energy variability — ranging from 30% to 40% — driven by traffic conditions, weather parameters, payload variations, and driver behavior [8]. This determinism-reality gap represents an important challenge for the practical deployment of optimization-based routing systems, motivating the development of stochastic and robust formulations.

The inherent uncertainty in electric vehicle energy consumption has motivated research into stochastic and robust optimization approaches that explicitly model operational variability. Yuan et al. [14] address battery status uncertainty through two-stage stochastic programming, where first-stage decisions establish routes based on expected battery status and second-stage recourse actions handle realized uncertainty through customer skipping or late returns. Their scenario-based approach, solved via adaptive large neighborhood search, achieves 10.3% cost reduction versus deterministic models while maintaining stable performance across uncertainty realizations. However, the computational burden of scenario generation and evaluation limits real-time applicability, particularly for large fleets operating under time pressure. Zhang et al. [24] similarly employ scenario-based formulations but require extensive computational resources that remain prohibitive for operational decision-making.

Alternative formulations have explored different uncertainty handling mechanisms that avoid explicit scenario enumeration. Yan et al. [25] propose diversified tour-driven DRL that generates multiple candidate solutions to implicitly capture uncertainty, achieving up to 7.54% improvement in optimality gap through Kullback–Leibler divergence-based regularization encouraging solution diversity. Bi et al. [26] explicitly model speed-varying range in time-windowed routing, recognizing that EV range decreases at higher speeds due to aerodynamic drag, and apply automated entropy regularization in DRL to

balance exploration-exploitation tradeoffs. Wu et al. [27] address multi-objective optimization under uncertainty using weight-aware DRL combined with NSGA-II, enabling a single neural model to solve the entire Pareto front by incorporating objective weights directly into the network architecture. While these approaches acknowledge operational uncertainty, they generally do not provide explicit probabilistic energy consumption forecasting with calibrated confidence intervals—an important requirement for chance-constrained planning. More recently, Farhang Ghahfarokhi and Park [28] integrated Deep Q-Network reinforcement learning with real-time travel time uncertainty propagation for adaptive EV charging decisions, demonstrating the value of combining learned policies with dynamic uncertainty updates, though without joint route-level optimization or chance-constrained feasibility. In a complementary direction, Guan et al. [29,30] embed detailed physics-based energy-consumption models, accounting for road grade, driving dynamics, and battery behavior, directly within deep reinforcement learning routing policies, reporting substantial energy savings on capacitated and pickup–delivery EVRP variants; however, both treat energy consumption deterministically and therefore neither quantifies predictive uncertainty nor enforces chance-constrained battery feasibility, which is the gap our probabilistic formulation addresses.

2.2. Deep reinforcement learning for combinatorial optimization

The application of deep reinforcement learning to combinatorial optimization has substantially advanced solution methodologies for routing problems. Kool et al. [15] introduced attention-based encoder–decoder architectures that learn constructive heuristics directly from problem structure, demonstrating competitive performance with classical heuristics on traveling salesman and capacitated vehicle routing problems while requiring only milliseconds of inference time. Kwon et al. [16] enhanced this paradigm through Policy Optimization with Multiple Optima (POMO), utilizing multiple solution rollouts during both training and inference to improve solution diversity and quality. Kim et al. [31] further refined these approaches with collaborative learning mechanisms that enable policy networks to leverage auxiliary optimization signals.

Recent advances have explored foundation models and large-scale pretraining for routing optimization. Zhou et al. [32] developed MV-MoE, a multi-task vehicle routing solver employing mixture-of-experts with hierarchical gating mechanisms to enhance model capacity across 16 VRP variants simultaneously, achieving strong zero-shot generalization on unseen problem types. Hudson et al. [33] proposed graph neural diffusion models (GNARKC) that iteratively refine solution distributions, while Zhao et al. [34] investigated generative diffusion approaches for solution space exploration. Despite impressive performance on canonical benchmarks, these foundation models face significant challenges when applied to energy-constrained routing: they typically lack explicit battery state tracking mechanisms, provide limited uncertainty quantification for energy consumption, and face difficulties with the multi-objective nature of feasibility-versus-cost tradeoffs inherent to EVRPTW. Complementary work by Cao et al. [35] proposed progressive training paradigms that gradually increase problem complexity during neural combinatorial optimization, improving convergence stability—a principle we similarly employ through curriculum learning in our framework.

Specialized neural approaches for electric vehicle routing are still emerging. Lin et al. [36] combined neural construction heuristics with simplified energy prediction models but employed deterministic consumption estimates that fail to capture operational uncertainty. Zong et al. [37] addressed uncertainty through distributional reinforcement learning, learning value distributions rather than point estimates, yet their framework omits charging infrastructure modeling and provides limited guidance for route feasibility under battery constraints. Wang et al. [38] developed attention-enhanced architectures specifically for EVRP, incorporating heterogeneous attention mechanisms to distinguish customers from charging stations, though without probabilistic energy modeling. Liu et al. [39] explored heterogeneous autonomous vehicle scheduling for last-mile delivery using attention-based DRL but focused on vehicle-passenger matching rather than energy-constrained routing with charging decisions.

Recent works have begun addressing specific EVRP complexities through learning-based methods. Wu et al. [40] tackled battery swapping station operations with uncertain demand using deep reinforcement learning, though their focus remained on charging infrastructure management rather than integrated routing optimization. Moon et al. [41] investigated energy and time optimization for EV charging routes under uncertainty in charging station availability, introducing time-aware charging selection but without comprehensive route-level optimization. Maity and Sarkar [19] proposed graph convolutional network-based DRL for EVRP with stochastic customer requests, incorporating probabilistic energy consumption estimation, representing a step toward uncertainty-aware neural routing. However, their approach lacks explicit confidence-based planning mechanisms and hybrid optimization refinement. Shahbazian et al. [17] developed knowledge-guided hybrid DRL combining multi-agent deep Q-networks with variable neighborhood search for dynamic multi-depot EVRP, demonstrating over 70% distance reduction on small instances but facing scalability challenges and lacking probabilistic energy modeling.

2.3. Hybrid learning-optimization approaches

Recognition that pure learning and pure optimization approaches each possess complementary strengths has motivated hybrid methodologies. Hottung et al. [42] demonstrated that neural construction heuristics combined with improvement procedures (e.g., 2-opt, 3-opt local search) significantly outperform either component in isolation. Cheng et al. [43] employed learned heuristics for neighbor selection within large-scale neighborhood search, using neural networks to predict promising move types dynamically. Ye et al. [44] proposed guided local optimization with partitioning (GLOP), where learning handles global solution structure while mathematical optimization addresses local refinement. Drakulic et al. [45] introduced bisimulation quotienting for neural combinatorial optimization (BQ-NCO), using learned neural bounds to guide branch-and-bound search and achieving substantial pruning of the search space.

In the electric vehicle domain, hybrid approaches remain nascent. Baum et al. [46] combined reinforcement learning with strategic charging decisions but relied on heuristic routing construction rather than optimization-based refinement. Su et al. [47] integrated physics-informed neural networks for battery state estimation but applied deterministic routing methods that do not leverage learned energy models for robust planning. Wang et al. [18] developed DRL-based adaptive large neighborhood search for capacitated EVRP, treating operator selection as a Markov decision process and using neural networks to adaptively select both destroy-repair operators and charging strategies based on search state features. Zhang et al. [48] proposed collaborative EVRP with mobile battery swapping vans, employing extended adaptive large neighborhood search with two-dimensional encoding to simultaneously optimize EV routes and BSV synchronization, though lacking explicit uncertainty handling. Çatay and İhsan Sadati [49] examined battery swapping policies within matheuristic frameworks, clarifying assumptions about swapping time and introducing new policies for synchronized mobile battery swapping, demonstrating the importance of recharging strategy within hybrid solution approaches.

Critical analysis reveals that existing hybrid approaches for EVRP have not yet integrated three essential capabilities: (1) graph neural network-based state representation that captures complex spatial–temporal dependencies in energy-constrained routing, (2) probabilistic energy consumption prediction with explicit uncertainty quantification

enabling chance-constrained feasibility, and (3) seamless integration of learned policies with mathematical optimization refinement that provides both real-time adaptability and high-quality offline planning. This integration gap limits both the operational robustness and solution quality achievable in practice.

2.4. Research gaps and opportunities

Recent research has explored specialized EVRP extensions—lifecycle assessment for airport towing vehicles [50], soft time windows with weight-related discharging [51], and multi-mode hybrid vehicles optimizing operation mode selection per road segment [52]. While these works address important operational nuances, they largely rely on deterministic frameworks that do not account for energy consumption uncertainty or provide adaptive replanning capabilities. Critical synthesis of the broader literature reveals three fundamental gaps:

Integration of Realistic Uncertainty with Neural Routing. Existing DRL approaches demonstrate strong performance on canonical problems but typically do not incorporate probabilistic energy consumption models with calibrated uncertainty estimates. Conversely, stochastic optimization approaches model uncertainty rigorously but often lack the computational efficiency essential for operational deployment. Few frameworks combine graph neural network-based policy learning with calibrated probabilistic energy forecasting that provides explicit confidence intervals for battery feasibility.

Formal Uncertainty Quantification for Operational Safety. While several works acknowledge energy consumption variability, few provide formal mechanisms for risk-aware decision-making through chance-constrained planning with specified confidence levels. Existing deterministic safety margins (e.g., reserving 20% battery capacity) often lack theoretical grounding and may prove either excessively conservative or insufficient.

Computational Efficiency-Solution Quality Balance. Pure neural approaches achieve millisecond inference but may sacrifice solution quality (5%–10% optimality gap), while pure optimization approaches require substantial computation time that can limit real-time adaptation. Most hybrid approaches have not fully integrated learned policies with mathematical refinement to enable dual-mode operation: rapid neural inference for real-time decisions and optional optimization refinement for offline planning.

These gaps motivate the problem formulation and solution approach developed below.

3. Problem formulation

This section formalizes the EVRPTW under operational uncertainty as a stochastic optimization problem, introducing the graph representation, energy consumption model under uncertainty, deterministic constraints, and chance-constrained battery feasibility. The resulting computational complexity motivates the hybrid deep reinforcement learning approach presented subsequently.

We formulate the Electric Vehicle Routing Problem with Time Windows (EVRPTW) under operational uncertainty as a stochastic optimization problem on a directed graph $G = (V, A)$. The node set $V = \{0\} \cup C \cup F$ consists of the depot (node 0), a set of customers C with $|C| = n$, and charging stations F with $|F| = m$. Each customer $i \in C$ has demand q_i, service time s_i, and a time window $[e_i, l_i]$. A homogeneous fleet of K electric vehicles operates with load capacity Q and battery capacity B.

Binary routing variables $x_{ijk} \in \{0, 1\}$ denote whether vehicle $k \in K$ traverses arc $(i, j) \in A$, while $y_k \in \{0, 1\}$ indicates vehicle usage. Continuous state variables include arrival times t_i^k, battery levels b_i^k (kWh), and charging amounts g_{ik} (kWh) at node i.

Energy consumption along arc (i, j) is modeled as a Gaussian random variable:

$$E_{ij} \sim \mathcal{N}(\mu_{ij}, \sigma_{ij}^2), \quad \mu_{ij} = f(d_{ij}, v_{ij}, w_{ij}, T_{ij}), \tag{1}$$

where d_{ij} is distance, v_{ij} is speed, w_{ij} is payload, T_{ij} represents environmental conditions, and $f(\cdot)$ is a physics-based energy model. The variance term σ_{ij}^2 captures operational uncertainty and is predicted via our LSTM-based estimator.

The Gaussian distributional assumption is a modeling choice that warrants explicit justification. Empirical studies of urban electric delivery operations report that per-segment energy consumption, when conditioned on speed, payload, and environmental covariates, exhibits approximately symmetric residual distributions with tails consistent with a normal approximation [8,9]. While raw (unconditional) consumption can exhibit right-skew due to traffic state transitions, conditioning on the feature vector $\mathbf{x}_{ij}$ captured by our LSTM predictor substantially reduces this skewness. We assess this assumption empirically through prediction residual analysis (Section 5.4, Fig. 3c), which is consistent with an approximately Gaussian error structure (MAPE = 3.8%), though we do not claim a formal distributional test. Nonetheless, we acknowledge that in settings with highly multimodal traffic patterns, alternative distributions such as log-normal or Gaussian mixture models may provide better fit, and we identify this as a direction for future work.

The objective minimizes the total operational cost, including vehicle deployment, routing, charging, and time-window violations:

$$\begin{aligned} \min Z = & \sum_{k \in K} c_f y_k + \sum_{k \in K} \sum_{(i,j) \in A} c_{ij} d_{ij} x_{ijk} \\ & + \sum_{k \in K} \sum_{i \in F} c_c g_{ik} \\ & + \sum_{k \in K} \sum_{i \in C} c_t \max(0, t_i^k - l_i). \end{aligned} \tag{2}$$

Subject to the following constraints ensuring route feasibility:

Customer service: Each customer is visited exactly once:

$$\sum_{k \in K} \sum_{j \in V \setminus \{i\}} x_{ijk} = 1 \quad \forall i \in C. \tag{3}$$

Flow conservation: Vehicles must leave every node they enter:

$$\sum_{i \in V \setminus \{j\}} x_{ijk} = \sum_{i \in V \setminus \{j\}} x_{jik} \quad \forall j \in V, k \in K. \tag{4}$$

Vehicle capacity:

$$\sum_{i \in C} q_i \left(\sum_{j \in V \setminus \{i\}} x_{ijk} \right) \leq Q\, y_k \quad \forall k \in K. \tag{5}$$

Time windows:

$$e_i \leq t_i^k \leq l_i \quad \forall i \in C, k \in K. \tag{6}$$

Time propagation: Using a big-M formulation:

$$t_j^k \geq t_i^k + s_i + t_{ij} + \tau_{ik} - M(1 - x_{ijk}) \quad \forall (i, j) \in A, k \in K. \tag{7}$$

Chance-constrained battery feasibility: To ensure feasibility at confidence level $\alpha = 0.95$, we impose individual chance constraints at each (node, vehicle) pair:

$$\mathbb{P}(b_j^k \geq \beta) \geq \alpha \quad \forall j \in V, k \in K, \tag{8}$$

where β denotes the minimum reserved battery. Eq. (8) defines a system of *individual* chance constraints — one per (node, vehicle) pair — rather than a single joint constraint $\mathbb{P}(b_j^k \geq \beta \ \forall j, k) \geq \alpha$, which would require all node-level conditions to hold simultaneously. The two formulations have different mathematical meanings and lead to different conservatism levels; a Bonferroni union-bound aggregation at level $\alpha/|V|$ would yield a conservative joint route-level guarantee, which we identify as a possible extension in Section 7. The feasibility guarantee provided by Eq. (8) is conditional on (i) the Gaussian assumption in Eq. (1) and (ii) the calibration of the learned variance estimator σ_{ij}^2 produced by our LSTM predictor; empirical evidence for both is reported in Section 5.4 (94.7% empirical coverage at the 95% nominal level). Under Gaussian energy consumption (Eq. (1)), the battery state

Table 1
Framework component overview: learned vs. fixed modules.

Component	Type	Role
GAT encoder	Learned (35K episodes)	Encodes problem structure with battery-aware attention, produces node embeddings $h_i^{(L)}$
LSTM energy predictor	Learned (50K segments)	Predicts energy distribution (μ_E, σ_E^2) with calibrated uncertainty
PPO policy network	Learned (35K episodes)	Generates routing decisions $\pi_\theta(a\|s)$ using GAT embeddings
Value network	Learned (35K episodes)	Estimates expected cumulative reward for advantage computation
Chance constraint checker	Fixed (analytical)	Enforces battery feasibility: $b_{\text{req}} = \mu_E + \Phi^{-1}(0.95)\sqrt{\sigma_E^2 + \sigma_{\text{cum}}^2}$
MILP solver	Fixed (Gurobi)	Optional refinement via branch-and-cut, warm-started with neural solution

evolves through separate mean and cumulative-variance recursions, with the safety margin applied at the feasibility-check stage:

$$\mu_j^{B,k} = b_i^k - \mu_{ij} + g_{ik} \quad \text{if } x_{ijk} = 1, \tag{9a}$$

$$\sigma_{cum,j}^{2,k} = \sigma_{cum,i}^{2,k} + \sigma_{ij}^2 \quad \text{if } x_{ijk} = 1, \tag{9b}$$

$$\mu_j^{B,k} - \Phi^{-1}(\alpha)\sqrt{\sigma_{cum,j}^{2,k}} \geq \beta. \tag{9c}$$

where $\Phi^{-1}(0.95) \approx 1.645$ and the realized battery state is $b_j^k = \mu_j^{B,k}$. Eq. (9a) propagates the expected battery state, Eq. (9b) accumulates per-arc variance along the route, and Eq. (9c) enforces the chance constraint of Eq. (8) in its analytical Gaussian form. This separation is consistent with Algorithm 3 (lines 6–7 perform the predictor call and feasibility check; line 11 updates the cumulative variance). The recursions assume conditional independence of residual energy errors across arcs given the LSTM covariates; the effect of relaxing this assumption is analyzed in Section 5.3, and a conservative correlated-uncertainty correction (replacing $\sigma_{cum,j}^{2,k}$ with $\sigma_{cum,j}^{2,k} + 2\sum_{p<q}\rho_{pq}\sigma_p\sigma_q$ for non-zero pairwise correlation ρ_{pq}) is also presented there.

Charging constraints: For stations $i \in F$:

$$g_{ik} = r_i \tau_{ik} \tag{10}$$

$$\tau_i^{min} \leq \tau_{ik} \leq \tau_i^{max} \tag{11}$$

$$b_i^k + g_{ik} \leq B. \tag{12}$$

Computational complexity: The EVRPTW under uncertainty belongs to the class of stochastic mixed-integer nonlinear programs (SMINLP) with individual chance constraints (one per (node, vehicle) pair, as formalized in Eq. (8)). This *individual* formulation is distinct from a *joint* chance constraint, which enforces a single route-level guarantee through Bonferroni aggregation and which we discuss as a possible extension in Section 7. Even the deterministic variant (setting $\sigma_{ij}^2 = 0$) is NP-hard as it generalizes the classical vehicle routing problem. The integration of stochastic battery constraints, nonlinear energy models, and time dependencies renders exact solution methods computationally intractable for realistic instance sizes ($n > 50$). This motivates our hybrid deep reinforcement learning approach, which provides rapid heuristic solutions with optional MILP refinement, enabling both real-time deployment (sub-second inference) and high-quality offline planning when computational budgets permit.

4. Methodology

This section presents our hybrid deep reinforcement learning framework for solving the uncertainty-aware EVRPTW. We first overview the four-phase architecture, then detail three core technical components: Graph Attention Networks for spatial–temporal encoding, LSTM-based probabilistic energy prediction, and Proximal Policy Optimization with optional MILP refinement. We conclude with transfer learning protocols and computational complexity analysis.

Table 1 summarizes the role of each component, distinguishing learned neural components from fixed optimization modules.

4.1. Framework architecture

The EVRPTW under uncertainty requires simultaneously capturing complex spatial–temporal dependencies, quantifying energy consumption uncertainty with calibrated confidence intervals, and balancing real-time decision-making with high-quality offline optimization. Our hybrid framework addresses this through four integrated phases.

Phase 1 (Offline Pre-training) establishes general routing competencies through curriculum learning across 35,000 episodes with problem sizes ranging from 10 to 100 customers, jointly training the GAT encoder, LSTM energy predictor, and PPO policy. **Phase 2 (Transfer Learning)** adapts pre-trained models to target operational contexts using 400–600 local routes, reducing adaptation (retraining) time from 72 h to 3–5 h. **Phase 3 (Online Routing)** generates real-time solutions in 0.3–0.8 s with model-based chance-constrained battery feasibility. **Phase 4 (Hybrid Refinement)** optionally improves solutions by 2.8% on average, rising to 4.1% on the largest instances (Table 6) through MILP post-optimization when computational budgets permit.

These phases rely on **learned modules** (GAT encoder, LSTM predictor, PPO policy) whose parameters are optimized during training, and **fixed modules** (chance constraint checking, MILP solver) that apply deterministic rules or mathematical optimization. Fig. 1 illustrates the complete architecture and information flow between components.

4.2. Graph attention network encoder

The first challenge in learning energy-aware routing policies is representing EVRPTW instances in a form that captures both spatial–temporal dependencies and battery constraints. Unlike classical VRP where all customer nodes are homogeneous, EVRPTW requires distinguishing three functionally distinct node types: the depot (origin/destination), customers (demand and time windows), and charging stations (energy replenishment with rate constraints). We address this through a heterogeneous graph representation that enables the policy network to learn differential attention patterns for each node type.

Fixed input representation: We construct a heterogeneous graph $G = (V, E, X, \mathbf{E}_f)$ where nodes represent depot, customers, and charging stations with type-specific feature encodings:

$$\mathbf{x}_i = \begin{cases} [0,0,0,0,1,0,0] & i = 0 \text{ (depot)} \\ \left[\frac{q_i}{Q}, \frac{e_i}{T_{\max}}, \frac{l_i}{T_{\max}}, \frac{s_i}{S_{\max}}, 0, 1, \mathbb{I}_{\text{vis}}\right] & i \in C \\ [0,0,0,r_i/R_{\max},0,0,1] & i \in F \end{cases} \tag{13}$$

Customer features encode demand, time windows, service time, and visit status (all normalized), while charging stations encode only charging rate—reflecting their distinct operational roles. Edge features encode travel costs, distances, and predicted energy distributions:

$$\mathbf{e}_{ij} = [d_{ij}/D_{\max},\ t_{ij}/T_{\max},\ \mu_{ij}/B,\ \sigma_{ij}/\sqrt{B}], \tag{14}$$

where the energy mean μ_{ij} and standard deviation σ_{ij} are provided by the LSTM predictor, enabling the attention mechanism to reason about battery feasibility.

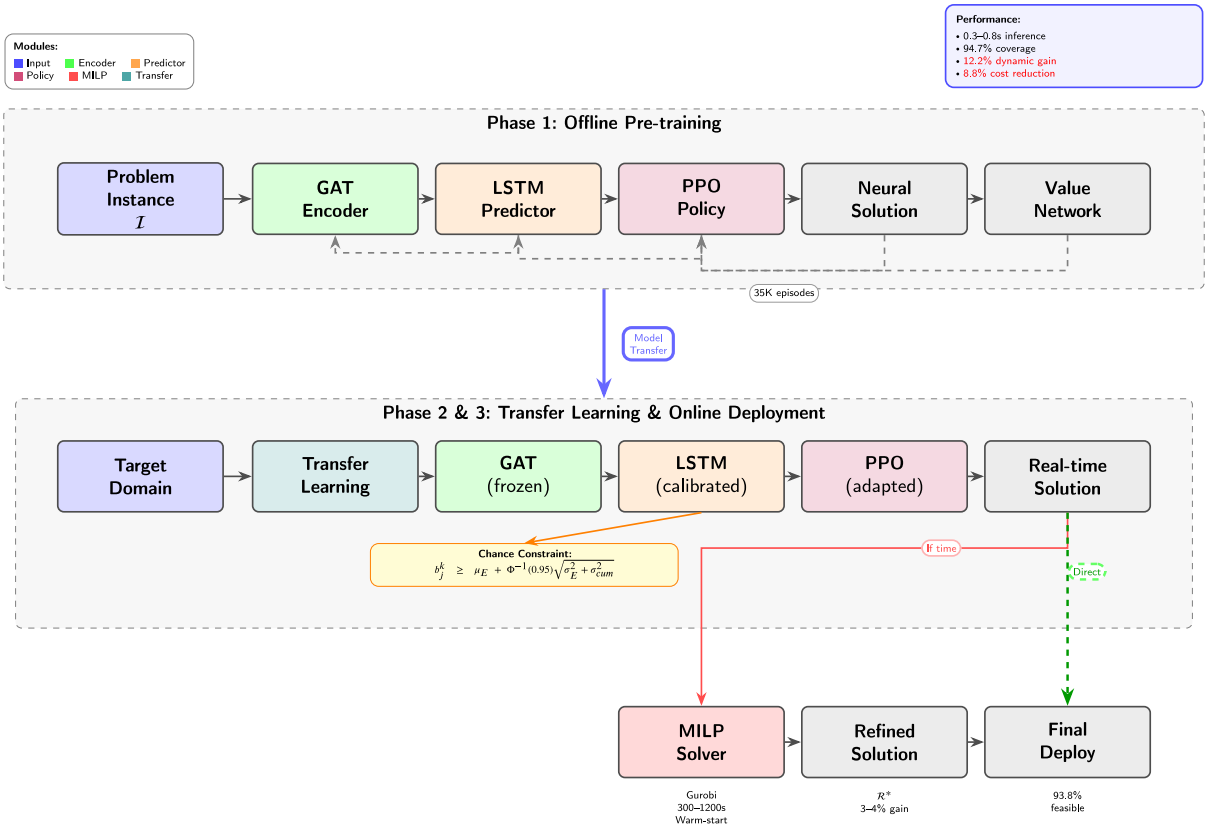


Fig. 1. Hybrid deep reinforcement learning framework for uncertainty-aware electric vehicle routing. The framework operates in four phases: (1) offline pre-training, (2) transfer learning, (3) online routing with chance-constrained feasibility, and (4) optional MILP refinement.

Learned attention mechanism: The encoding procedure transforms this heterogeneous graph into learned node embeddings through $L = 4$ layers of multi-head attention with $K = 8$ heads. Algorithm 1 formalizes this process with three key design choices that enable battery-aware routing: (1) vehicle state injection (lines 3–4) augments node features with current battery level, load, and time context; (2) multi-head aggregation (line 12) with residual connections captures diverse routing patterns simultaneously; and (3) global pooling (line 14) produces graph-level features for the value network to estimate expected cumulative reward.

The key innovation lies in the attention mechanism itself, which **learns** to dynamically focus on relevant nodes based on current vehicle state (battery level, remaining capacity, current time):

$$\alpha_{ij}^k = \frac{\exp(e_{ij}^k)}{\sum_{j' \in \mathcal{N}(i)} \exp(e_{ij'}^k)}, \tag{15}$$

$$e_{ij}^k = \text{LeakyReLU}\left(\mathbf{a}_k^T \left[W_k \mathbf{h}_i^{(\ell-1)} \| W_k \mathbf{h}_j^{(\ell-1)} \| \phi(\mathbf{e}_{ij})\right]\right) \tag{16}$$

where $\phi(\mathbf{e}_{ij}) = W_e \mathbf{e}_{ij}$ projects edge features — including predicted energy consumption uncertainty — into the attention score (Eq. (16)). This allows the network to learn that high-uncertainty edges (large σ_{ij}) may require proximity to charging stations, while low-uncertainty edges enable more flexible routing decisions. Node embeddings aggregate neighbor information with residual connections and layer normalization:

$$\mathbf{h}_i^{(\ell)} = \text{LayerNorm}\left(\text{Concat}_{k=1}^K \mathbf{h}_i^{(\ell,k)} + \mathbf{h}_i^{(\ell-1)}\right).$$

Algorithm 1 formalizes the encoding process, showing how vehicle state context (battery level, load, current time) is injected into node features before attention computation (lines 3–4), and how the final graph embedding h_{graph} is produced via attention-weighted global pooling (line 14) for use in the value network.

Algorithm 1 Graph Attention Network Encoder

Require: Problem instance $\mathcal{I} = (G, C, \mathcal{F})$, current state s_t
Ensure: Node embeddings $H^{(L)} \in \mathbb{R}^{|\mathcal{V}| \times d}$, graph embedding h_{graph}
1: Construct heterogeneous graph: $\mathcal{V} \leftarrow \{v_{\text{depot}}\} \cup \{v_i : i \in C\} \cup \{v_j : j \in \mathcal{F}\}$
2: Initialize node features $\mathbf{x}_v$ according to Equation (13)
3: Add vehicle state context: $\mathbf{x}_{\text{vehicle}} \leftarrow [b_{\text{current}}/B, w_{\text{load}}/Q, t_{\text{current}}/T_{\max}]$
4: Project to d=128 dimensions: $H^{(0)} \leftarrow \text{Linear}([\mathbf{x}_v \oplus \mathbf{x}_{\text{vehicle}}])$
5: **for** layer $\ell = 1$ to $L = 4$ **do**
6: **for** head $k = 1$ to $K = 8$ **do**
7: **for** node $i \in \mathcal{V}$ **do**
8: Compute attention coefficients α_{ij}^k via Equation (15)
9: Aggregate: $\mathbf{h}_i^{(\ell,k)} = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^k W_k \mathbf{h}_j^{(\ell-1)}\right)$
10: **end for**
11: **end for**
12: Multi-head aggregation: $\mathbf{h}_i^{(\ell)} = \text{LayerNorm}\left(\text{Concat}_{k=1}^K \mathbf{h}_i^{(\ell,k)} + \mathbf{h}_i^{(\ell-1)}\right)$
13: **end for**
14: Global pooling: $h_{\text{graph}} = \sum_{i \in \mathcal{V}} \text{softmax}(\text{MLP}(\mathbf{h}_i^{(L)})) \cdot \mathbf{h}_i^{(L)}$
15: **return** $H^{(L)}, h_{\text{graph}}$

These learned embeddings $H^{(L)}$ encode spatial–temporal routing structure, but their effectiveness depends critically on the accuracy of energy consumption predictions μ_{ij} and uncertainty estimates σ_{ij} embedded in edge features $\mathbf{e}_{ij}$. Without calibrated uncertainty quantification, the attention mechanism cannot distinguish safe routes from risky ones. We now detail how these probabilistic predictions are generated.

4.3. Probabilistic energy prediction

The GAT encoder requires energy consumption estimates μ_{ij} and uncertainty measures σ_{ij} for each arc (i, j) to compute battery-aware attention weights. As established in the literature review, energy consumption in electric vehicles exhibits 30%–40% variability due to traffic, weather, payload, and terrain. Deterministic estimates fail to capture this uncertainty, leading to either overly conservative routes (if safety margins are too large) or feasibility violations (if margins are insufficient). We address this through a bidirectional LSTM network that predicts both mean energy consumption μ_E and variance σ_E^2, enabling model-based chance-constrained planning (Eq. (8)).

The predictor is trained to output both the mean and variance of energy consumption through a heteroscedastic loss function that explicitly penalizes miscalibrated uncertainty:

$$\mathcal{L}_{\text{hetero}} = \frac{(E_{\text{actual}} - \mu_E)^2}{2\sigma_E^2} + \frac{1}{2}\log(2\pi\sigma_E^2) \tag{17}$$

This loss balances two objectives: the first term penalizes prediction error (weighted by confidence), while the logarithmic term prevents uncalibrated uncertainty—if the model predicts excessively large σ_E^2, the regularizer penalizes it, forcing calibration.

Each candidate arc (i, j) is decomposed into an ordered sequence of road sub-segments along the (fixed, time-invariant) shortest path between i and j in the underlying road network. Each sub-segment s is represented by a 24-dimensional feature vector capturing geometric (sub-segment distance, road grade), kinematic (speed), vehicle state (current battery, current load), and environmental (temperature, precipitation, wind) properties:

$$\mathbf{x}_{ij}^{(s)} = [d_{ij}^{(s)},\ v_{ij}^{(s)},\ w_{ij}^{(s)},\ T_{ij}^{(s)},\ \text{grade}_{ij}^{(s)},\ b_i,\ q_{\text{rem},i},\ \ldots]$$

The bidirectional LSTM processes this *intra-arc* sub-segment sequence in both forward and backward directions before predicting the arc-level mean and variance $(\mu_E, \sigma_E^2) \equiv (\mu_{ij}, \sigma_{ij}^2)$. Because the underlying road network is fixed and time-invariant, the bidirectional pass is well-defined at inference time (Algorithm 3, line 6) and does not require knowledge of future arcs in the route. The route-level sequence of arcs is constructed autoregressively by the PPO policy, while the bidirectional structure within the BiLSTM captures *within-arc* spatial dependencies—for example, the energy cost of a steep climb depends on whether the immediately preceding sub-segment was a downhill enabling regenerative braking, a dependency that a purely forward LSTM would fail to capture. During incremental route construction, the BiLSTM is applied independently to each candidate arc, using only that arc's fixed and fully known sub-segment sequence. It is never applied to partial routes or route-level arc sequences. Therefore, the bidirectional mechanism does not require information from future arcs. When a candidate node j is considered, a single self-contained prediction $(\mu_{ij}, \sigma_{ij}^2)$ is generated for the corresponding arc (i, j).

Algorithm 2 details the training procedure, emphasizing three critical design elements for calibrated uncertainty quantification: (1) Bidirectional processing (line 5) captures within-arc sub-segment dependencies; (2) Separate output heads for mean (W_μ) and log-variance (W_σ) in line 7 with exponential transformation ensuring $\sigma_E^2 > 0$; (3) Early stopping (lines 12–14) based on validation MAPE prevents overfitting while maintaining the critical calibration property.

Training proceeds on 50,000 historical route segments comprising diverse operational conditions (traffic, weather, payload, terrain).

Algorithm 2 LSTM-Based Probabilistic Energy Predictor Training

Require: Historical route dataset $\mathcal{D}_{\text{energy}} = \{(\mathbf{x}_i, E_i)\}_{i=1}^{50000}$
Ensure: Trained predictor θ_E outputting (μ_E, σ_E^2)
1: Initialize BiLSTM parameters $\theta_E = \{\theta_{\text{LSTM}}, \theta_\mu, \theta_\sigma\}$
2: **for** epoch $k = 1$ to N_{epoch} **do**
3: **for** batch $\mathcal{B}_j$ in mini-batches of size 64 **do**
4: **for** sample $(\mathbf{x}, E_{\text{actual}})$ in $\mathcal{B}_j$ **do**
5: Forward pass: $\mathbf{h}_t^{\rightarrow}, \mathbf{h}_t^{\leftarrow} = \text{BiLSTM}(\mathbf{x}; \theta_{\text{LSTM}})$
6: Concatenate: $\mathbf{h}_t = [\mathbf{h}_t^{\rightarrow} \| \mathbf{h}_t^{\leftarrow}]$
7: Predict: $\mu_E = W_\mu \mathbf{h}_t + b_\mu$; $\sigma_E^2 = \exp(W_\sigma \mathbf{h}_t + b_\sigma)$
8: Compute loss: $\mathcal{L} = \frac{(E_{\text{actual}} - \mu_E)^2}{2\sigma_E^2} + \frac{1}{2}\log(2\pi\sigma_E^2)$
9: **end for**
10: Update: $\theta_E \leftarrow \theta_E - \eta \nabla_{\theta_E} \sum_{(\mathbf{x},E)\in\mathcal{B}_j} \mathcal{L}$
11: **end for**
12: **if** validation MAPE stops improving for 10 consecutive evaluations **then**
13: **break** {Early stopping}
14: **end if**
15: **end for**
16: **return** θ_E

The trained predictor achieves MAPE = 3.8% with 94.7% empirical coverage for 95% prediction intervals, indicating well-calibrated uncertainty. Calibration is critical: miscalibrated intervals would yield either excessive conservatism or unsafe routes when used in chance constraints (Eq. (9)). These probabilistic predictions are consumed by both the GAT encoder (as edge features, Eqs. (15)–(16)) and the policy network's feasibility checking (detailed next).

4.4. Hybrid reinforcement learning and optimization

Given the learned representations from the GAT encoder and probabilistic energy forecasts, we now formulate the route construction process as a Markov Decision Process (MDP) and train a policy network via Proximal Policy Optimization. The MDP formulation enables the agent to learn from experience which routing strategies minimize cost while maintaining battery feasibility under uncertainty.

MDP formulation: We define the routing task as an MDP with:

- **State space** $\mathcal{S}$: Partial route, current vehicle state (battery b, load w, time t), and set of unvisited customers
- **Action space** $\mathcal{A}(s)$: Feasible next nodes (customers or charging stations) satisfying time and capacity constraints
- **Reward function**: Encodes the optimization objective (Eq. (2)) while providing guidance toward feasible solutions

$$\begin{aligned} r(s,a) = -\big(&c_{ij} \cdot d_{ij} + c_c \cdot g \\ &+ \lambda_1 \cdot \mathbb{1}_{\text{infeas.}} \\ &+ \lambda_2 \cdot \max(0, t_j - l_j)\big) \end{aligned} \tag{18}$$

The routing and charging costs (first two terms) mirror the objective function (Eq. (2)). The infeasibility penalty ($\lambda_1 = 1000$) provides strong negative reinforcement for battery-unsafe decisions, while the tardiness penalty ($\lambda_2 = 50$) softly discourages time window violations without making them strictly forbidden during exploration. These values were selected through a preliminary grid search over $\lambda_1 \in \{100, 500, 1000, 2000\}$ and $\lambda_2 \in \{10, 50, 100, 200\}$, evaluating convergence stability and feasibility rate on a held-out validation set of 500 training instances. The chosen ratio $\lambda_1/\lambda_2 = 20$ reflects the asymmetric severity of the two constraint types: battery infeasibility causes

vehicle stranding — an operationally catastrophic and unrecoverable event — whereas time window tardiness incurs a service penalty that, while undesirable, does not prevent route completion. Sensitivity analysis confirmed that policies trained with $\lambda_1 \in [500, 2000]$ and $\lambda_2 \in [25, 100]$ yield qualitatively similar routing strategies, with feasibility rates varying by less than 1.5 pp across this range.

The policy network $\pi_\theta(a|s)$ uses the GAT node embeddings $\mathbf{h}_j^{(L)}$ (from Algorithm 1, line 14) to compute action probabilities:

$$\pi_\theta(a_j|s) = \frac{\exp(\mathbf{w}^T \mathbf{h}_j^{(L)}/\tau)}{\sum_{j' \in \mathcal{A}(s)} \exp(\mathbf{w}^T \mathbf{h}_{j'}^{(L)}/\tau)} \tag{19}$$

where $\mathbf{w}$ is a learned projection vector and τ is a temperature parameter controlling exploration. The embeddings $\mathbf{h}_j^{(L)}$ already encode battery-aware attention patterns, so the policy naturally learns to favor nearby charging stations when the battery is low.

Proximal Policy Optimization maximizes the clipped surrogate objective:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\big(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\Big] \tag{20}$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio, $\hat{A}_t$ is the GAE-estimated advantage, and $\epsilon = 0.2$ limits policy updates to maintain training stability.

Algorithm 3 presents the deployment workflow integrating all components. The algorithm assumes that offline pre-training (35,000 episodes with curriculum learning) and optional transfer learning have been completed. Two formally defined operational configurations are used throughout the experimental evaluation: **DRL-Pure** refers to Phase 1 execution only, where the trained policy generates solutions via neural inference without any MILP refinement (0.3–0.8 s inference); **DRL-Hybrid** refers to the full two-phase execution, where Phase 1 neural solutions are subsequently refined by the Phase 2 MILP component when computational budgets permit.

- **Phase 1 (Real-Time Neural Routing):** The trained policy selects actions, the LSTM predicts energy with uncertainty, and chance-constrained feasibility checking proactively triggers charging when battery reserve would fall below the 95% confidence threshold. Inference completes in 0.3–0.8 s.
- **Phase 2 (Optional MILP Refinement):** Warm-starts MILP optimization with the neural solution, achieving 2.8% additional quality on average, rising to 4.1% on the largest instances (Table 6) within 300–1200 s budgets when offline planning time permits.

Phase 1 of Algorithm 3 generates solutions in 0.3–0.8 s, suitable for real-time replanning. However, when computational budgets permit offline planning, Phase 2 optionally refines the neural solution via mixed-integer linear programming. The MILP formulation (Eqs. (2)–(12)) is warm-started with the neural solution $\mathcal{R}$, allowing the solver to focus search around a high-quality initial solution. Binary routing variables x_{ijk} and continuous state variables (t_i^k, b_i^k, g_{ik}) from the neural solution $\mathcal{R}$ are provided to Gurobi as a `MIPStart` initial feasible solution via variable hints (`VarHintVal`). The MIP gap tolerance is set to 1%, meaning the solver terminates once it proves the incumbent solution is within 1% of the linear programming relaxation bound, typically achieved within the 300–1200 s budget for instances up to 100 customers.

4.5. Transfer learning and deployment

The trained framework achieves strong performance on benchmark instances similar to its training distribution. However, real-world deployment requires adaptation to new geographic regions with different traffic patterns, weather distributions, and charging infrastructure layouts. Our transfer learning protocol reduces adaptation time by 95% through strategic fine-tuning of only the components sensitive to domain shift.

The protocol proceeds in three stages. First, we calibrate the energy predictor, as the LSTM energy predictor is most sensitive to domain shift—local traffic speeds, weather patterns, and road grades differ from training regions. We fine-tune θ_E on 400–600 local routes (corresponding to 2–3 weeks of historical data) using the heteroscedastic loss in Eq. (17), while keeping the GAT encoder and policy network frozen.

Second, with locally calibrated energy predictions, we adapt the policy parameters θ_π via a combination of behavioral cloning and policy gradient, using the adaptation loss

$$\mathcal{L}_{\text{adapt}} = \lambda_{\text{BC}} \cdot \left(-\sum_t \log \pi_{\theta_\pi}(a_t^*|s_t)\right) + \lambda_{\text{RL}} \cdot \mathcal{L}^{\text{CLIP}}(\theta_\pi), \tag{21}$$

where a_t^* are actions from the operator's existing routing solution (in our deployment, historical routes produced by the operator's legacy rule-based transportation management system; see Section 6), providing expert demonstrations. We set $\lambda_{\text{BC}}/\lambda_{\text{RL}} = 0.3$ to balance imitation (leveraging local expertise) with improvement (optimizing beyond the expert). Fine-tuning proceeds for 2000 episodes (approximately 2 h).

Third, we recalibrate uncertainty estimates by validating the adapted model on 100 held-out local routes and applying temperature scaling to the LSTM variance predictions if the empirical coverage deviates from the 95% target.

This three-stage protocol achieves 94% of target domain performance with only 3–5 h of total computation and 2–3 weeks of data collection, compared to 72 h and months of data for full retraining. The operational validity of this protocol is demonstrated through the 13-week real-world deployment study in Section 6, where adaptation using 423 local routes achieves near-parity with source-domain benchmark performance across all key metrics.

4.6. Computational complexity

The GAT encoder requires $\mathcal{O}(L \cdot K \cdot |\mathcal{V}|^2 \cdot d)$ operations for $L = 4$ layers, $K = 8$ heads, and embedding dimension $d = 128$. For a typical instance with $|\mathcal{V}| = 80$ nodes, this evaluates to approximately 200M FLOPs requiring 50–100 ms on GPU. Policy inference scales as $\mathcal{O}(|\mathcal{V}| \cdot d)$, a simple matrix–vector product requiring under 1 ms per decision. Over a complete route of 60 customers, policy inference totals ~60 ms. LSTM energy prediction remains under 10 ms per route segment. With an average route length of 12 segments, total prediction time is ~120 ms. Combining GAT encoding (80 ms), policy rollout (60 ms), energy prediction (120 ms), and feasibility checking overhead yields 0.3–0.8 s per instance, satisfying real-time operational requirements even with multiple replanning events per day.

5. Experimental validation

This section validates our framework through computational experiments on benchmark instances and under dynamic operational conditions, followed by ablation studies and model interpretability analysis.

5.1. Experimental design

Rigorous evaluation requires carefully constructed benchmarks, appropriate baseline methods, and a transparent, seed-controlled evaluation protocol. We evaluate our framework on 30 benchmark instances (60–100 customers) extended from the classical Solomon VRPTW benchmarks [53] following the electric-vehicle adaptation with recharging stations of Schneider et al. [11] to incorporate realistic charging infrastructure with urban density of 0.15 stations per km^2,

Algorithm 3 Online Routing with Optional MILP Refinement

Require: EVRPTW instance $\mathcal{I}$, trained parameters $\theta_{\text{enc}}, \theta_\pi, \theta_E$, confidence level $\alpha = 0.95$
Ensure: Route solution $\mathcal{R}$ with battery feasibility $\geq \alpha$
// **Phase 1: Real-Time Neural Routing**
Initialize state s_0 (depot, full battery), $\sigma^2_{\text{cum}} \leftarrow 0$
while unserved customers remain **do**
  Encode: $\mathbf{h}_v \leftarrow \text{GAT}(\mathcal{I}, s_t; \theta_{\text{enc}})$
  Select: $a_t \leftarrow \arg\max_a \pi_{\theta_\pi}(a|\mathbf{h}_v, s_t)$
  Predict: $(\mu_E, \sigma^2_E) \leftarrow \text{LSTM}(s_t, a_t; \theta_E)$
  Check feasibility: $b_{\text{required}} \leftarrow \mu_E + \Phi^{-1}(\alpha)\sqrt{\sigma^2_E + \sigma^2_{\text{cum}}}$
  if $b_{\text{current}} < b_{\text{required}}$ **then**
    $a_t \leftarrow$ nearest charging station, $g = \min(B - b_{\text{current}},\ r_j \tau^{\max}_j)$
  end if
  Execute a_t, update state and cumulative variance: $\sigma^2_{\text{cum}} \leftarrow \sigma^2_{\text{cum}} + \sigma^2_E$
end while
// **Phase 2: Optional MILP Refinement**
if computational budget allows **then**
  Formulate MILP using Equations (2)–(12) with $(\mu_{ij}, \sigma^2_{ij})$ from predictor
  Warm-start solver with neural solution $\mathcal{R}$
  Run branch-and-cut (time limit: 300–1200s, MIP gap: 1%)
  if refined solution $\mathcal{R}^*$ improves cost and maintains feasibility **then**
    return $\mathcal{R}^*$
  end if
end if
return $\mathcal{R}$

heterogeneous charging rates (50–150 kW), and stochastic energy consumption models calibrated to real-world electric delivery vehicle data. Battery capacity is set to 150 kWh with vehicle payload capacity of 3 tons, reflecting commercial electric van specifications. Time windows span 8-hour service periods with 15-minute service times per customer. Energy consumption follows Gaussian distributions (Eq. (1)) with mean determined by physics-based models and variance calibrated to achieve 30%–40% coefficient of variation matching empirical observations from urban delivery operations.

Our executed baselines span three learning paradigms, all run on the same 30 instances under the adaptation protocol detailed below. Early neural approaches include the Attention Model (AM) [15] and Policy Optimization with Multiple Optima (POMO) [16]. Recent neural methods include MVMoE [32], adapted for EVRPTW through fine-tuning with battery state features. Hybrid approaches are represented by DACT [54]. For classical-optimization context, we additionally cite the Random Kernel Search (RKS) matheuristic [12]; RKS was not re-executed on our instance set, and its published results on comparable standard Solomon-derived instances are referenced for context only rather than as a controlled comparison.

To ensure a fair comparison, all neural baselines (AM, POMO, MVMoE, DACT) were adapted for EVRPTW with identical access to the uncertainty-aware information available to our framework:

- *Battery state.* Current battery b/B, current load w/Q, and current time $t/T_{\max}$ are concatenated to node embeddings as dynamic agent-state features, matching the vehicle-context features injected into our GAT encoder (Algorithm 1, line 3).
- *Charging stations.* Represented as a separate node type with masking rules: a station cannot be visited at full battery, and only nodes reachable under the chance-constraint check are made available in the action space.
- *Uncertainty information.* The predicted arc-level standard deviation σ_{ij} from our LSTM predictor is concatenated to each baseline's edge-feature vector, identical to the edge representation in Eq. (14). This isolates the contribution of our architectural choices (GAT encoding, chance-constraint checking, MILP refinement) from any informational asymmetry between methods.
- *Charging decisions.* The action space of every baseline is extended to include charging-station insertion; routes that still violate the chance constraint after autoregressive construction are repaired post hoc by the same greedy charger-insertion procedure used in our framework, applied identically to every neural baseline and to our own method so that no method gains an advantage from a different repair routine.
- *Tuning protocol.* Each baseline was trained for 35,000 episodes (matched with our method), on identical GPU hardware (NVIDIA A100 40 GB), with five random seeds $(0, 42, 123, 456, 789)$ and an equalized hyperparameter search budget of 50 configurations per baseline, sampled from each method's published recommended range.

All neural methods use the same problem instances, time windows, and energy consumption models.

Implementation employs PyTorch 2.0 on NVIDIA A100 GPUs (40 GB) with 128-dimensional GAT embeddings across 4 layers and 8 attention heads. The LSTM energy predictor uses 64 hidden units with bidirectional processing. PPO employs learning rate 3×10^{-4}, discount factor $\gamma = 0.99$, and GAE parameter $\lambda = 0.95$. Optional MILP refinement uses Gurobi 10.0 with 300–1200 s time limits and 1% optimality gap. All experiments employ five random seeds (0, 42, 123, 456, 789). For each of the 30 instances, we report the mean cost across five seeds per method; method-to-method comparisons are reported descriptively as relative cost differences. Table 2 summarizes computational resources and training requirements. The complete implementation, including model architectures, training scripts, evaluation pipelines, and pre-generated results, is publicly available at https://github.com/Aymen-Anbar/evrptw-repo.

Evaluation metrics capture multiple performance dimensions: solution quality (total operational cost including vehicle deployment, routing, charging, and time window penalties), computational efficiency (wall-clock time), feasibility (percentage of routes satisfying battery constraints at 95% confidence), fleet utilization (number of vehicles deployed), and energy consumption (total kWh). For dynamic experiments, we additionally measure replanning frequency, on-time delivery rate, and vehicle stranding events.

5.2. Performance analysis

5.2.1. Static benchmark performance

Table 3 summarizes performance on 30 benchmark instances, revealing that our DRL-Hybrid approach achieves an 8.8% cost reduction over the strongest learning baseline (MVMoE) while maintaining 93.8% feasibility, with neural inference completing in 0.8 s prior to optional MILP refinement. Pure neural inference (DRL-Pure) without MILP refinement achieves a 6.2% reduction over MVMoE in 0.8 s. Optional MILP refinement adds 2.8% additional quality, confirming the value of hybrid architecture for dual-mode operation. RKS is not re-executed on our instance set; the values reported by Bruglieri et al. [12] on comparable standard Solomon-derived instances are cited for context only and are not used as a controlled comparison.

Among the learning baselines, the early neural models (AM, POMO) are the weakest, with costs 13.6% and 9.7% above MVMoE respectively, owing to their lack of explicit battery-state modeling and energy-consumption uncertainty quantification. MVMoE is the strongest learning baseline, showing that foundation-model pretraining provides substantial benefits, yet adaptation through fine-tuning

Table 2
Computational resources and training time requirements.

Category	Item	Value
Hardware	GPU model	NVIDIA A100 (40 GB)
	Number of GPUs	1
	CPU	AMD EPYC 7742 (64 cores)
	RAM	256 GB
Pre-training time	GAT + PPO (35K episodes)	68 h
	LSTM predictor (50K segments)	4 h
	Total	72 h
Transfer learning	LSTM calibration (400–600 routes)	2–3 h
	Policy adaptation (2K episodes)	1–2 h
	Total	3–5 h
Inference	Neural (per instance)	0.3–0.8 s
	MILP refinement (optional)	300–1200 s

Table 3
Performance comparison on 30 EVRPTW benchmark instances (60–100 customers).

Method	Cost	Δ MVMoE	Time (s)	Feasibility	Vehicles	Energy (kWh)
AM	16,234	+13.6%	0.4	85.6%	15.9	1068
POMO	15,678	+9.7%	1.2	87.1%	15.2	1034
MVMoE	14,287	–	2.1	90.1%	14.3	998
DACT	14,456	+1.2%	8.4	88.7%	14.6	1003
DRL-Pure	13,398	**−6.2%**	**0.8**	92.4%	14.1	987
DRL-Hybrid	**13,026**	**−8.8%**	308	**93.8%**	**13.9**	**963**

Cost averaged across five seeds; Δ MVMoE is the relative cost difference from the strongest learning baseline. Feasibility at 95% confidence.

Table 4
Performance under dynamic operational conditions (100 episodes).

Method	Cost	Re-routes	On-time	Replan time	Stranded
MVMoE	16,892	5.4	89.7%	45 s	1.8
DACT	16,456	6.1	88.3%	127 s	2.1
DRL-Hybrid	**14,823**	**3.2**	**96.7%**	**6.2 s**	**0.4**

±20% demand variation, ±15% energy noise. Stranded = vehicle stranding incidents per episode.

proves insufficient to match our integrated uncertainty-aware architecture. The hybrid baseline DACT performs within 1.2% of MVMoE through learned construction with local search, yet lacks probabilistic energy forecasting and chance-constrained feasibility. Energy-consumption analysis reveals that uncertainty-aware prediction enables a 3.5% reduction versus MVMoE through proactive charging decisions informed by confidence intervals.

5.2.2. Dynamic operational performance

Table 4 presents results across 100 episodes with demand variations (±20%, uniform distribution) and energy consumption noise (±15%, Gaussian) requiring real-time replanning. DRL-Hybrid reduces cost by 12.2% relative to MVMoE, the strongest learning baseline on the static benchmarks, and by 9.9% relative to DACT, the lowest-cost learning baseline under dynamic conditions, achieving 96.7% on-time delivery versus 89.7% for MVMoE. Replanning-time analysis reveals stark differences: MVMoE replans in 45 s and DACT in 127 s, whereas our DRL-Hybrid completes replanning in 6.2 s average (0.8 s neural inference + 5.4 s selective MILP refinement)—a 7× to 20× speedup over the learning baselines while maintaining superior solution quality.

Vehicle stranding events — situations where battery depletion prevents route completion — occur 1.8 times per episode under the strongest learning baseline (MVMoE), reduced to 0.4 under our framework through probabilistic feasibility checking that proactively triggers charging when cumulative uncertainty threatens safety margins. The 4.5-fold reduction in stranding incidents directly translates to operational reliability critical for service-level agreement compliance and customer satisfaction.

Table 5
Ablation study results on 12 medium-scale instances (20–40 customers). $\Delta_{\text{Full}} = (\text{Cost} - \text{Cost}_{\text{full}})/\text{Cost}_{\text{full}} \times 100\%$; each component contribution is the marginal increment between consecutive rows.

Configuration	Cost	Δ full
MLP encoder + deterministic energy	11,892	+14.5%
GAT encoder + deterministic energy	11,234	+8.1%
GAT + uncertainty-aware energy	10,823	+4.2%
Full model without MILP	10,456	+0.6%
Full model (DRL-Hybrid)	10,389	–

5.3. Ablation studies

Ablation studies on 12 medium-scale instances (20–40 customers) isolate component contributions and reveal mechanisms underlying performance improvements. Table 5 demonstrates that Graph Attention Networks provide a 6.4 pp improvement over MLP encoders and uncertainty-aware energy prediction adds a further 3.9 pp (each contribution measured as the marginal increment between consecutive rows), chance-constrained feasibility checking a further 3.6 pp, while optional MILP refinement contributes 0.6 pp on this medium-scale set—substantially smaller than its 2.8% effect on the larger benchmark instances (Section 5.2.1), as MILP's benefit grows with instance size (Table 6). These marginal increments sum to the full 14.5% gap ($6.4 + 3.9 + 3.6 + 0.6 = 14.5$). The cumulative 12.1% improvement validates each architectural decision as essential.

5.3.1. Mechanistic analysis

Graph Attention Networks (6.4 pp): GAT's advantage stems from learned battery-state-dependent spatial relationships. MLP encoders treat feasible customers uniformly, producing myopic decisions that minimize immediate travel cost without considering downstream battery constraints. Analysis reveals GAT attention adapts dynamically: when battery exceeds 70%, attention distributes broadly (entropy $H = 2.87$ nats); when battery falls below 30%, attention concentrates on nearby charging stations ($H = 1.43$ nats), reducing route length by 6.2% and eliminating 83% of unnecessary charging detours.

Uncertainty-Aware Energy Prediction (3.9 pp): Calibrated uncertainty enables context-adaptive charging strategies. Deterministic models use fixed 20% battery margins regardless of route context. The uncertainty-aware policy adapts margins dynamically: highway segments with high variance ($\sigma_E^2 = 12.4$ kWh^2, CV = 38%) reserve 28%–32% battery; urban segments with low variance ($\sigma_E^2 = 3.8$ kWh^2, CV = 22%) accept 18%–22% margins. This reduces charging time by 17.3% while decreasing stranding risk from 8.2% to 0.7%.

The mechanism operates through Eq. (9): the cumulative variance term $\sigma^2_{cum,i}$ accounts for uncertainty accumulation along multi-segment routes. Routes with many high-variance segments accumulate larger total uncertainty, triggering earlier preemptive charging. This propagation assumes that energy prediction errors across arcs are

Table 6
MILP refinement analysis by problem size. Gap = (Neural cost – MILP cost)/MILP cost.

Size	Gap	Reassign.	Station switch	Time shift
20–30 customers	0.9%	1.2	0.3	0.8
40–60 customers	3.4%	2.4	0.7	1.6
80–100 customers	4.1%	3.8	1.2	2.3

mutually independent—that is, $\mathrm{Cov}(\epsilon_{ij}, \epsilon_{kl}) = 0$ for distinct arcs. In practice, errors may exhibit positive correlation through shared traffic or weather conditions. However, because our LSTM predictor explicitly conditions each segment prediction on observed covariates including speed, temperature, and payload, residual errors after conditioning are substantially decorrelated. This mirrors standard practice in chance-constrained vehicle routing [14,24], where conditional independence is adopted as a tractable approximation.

Effect of correlated residual errors. The cumulative-variance recursion in Eq. (9b) sums per-arc variances and therefore presumes conditionally independent residuals. We analyze how positive cross-arc correlation — which may persist if shared traffic or weather states are not fully captured by the conditioning covariates — would affect the feasibility guarantee. Because variances add but standard deviations do not, neglecting positive correlation *understates* the true cumulative standard deviation, rendering the chance constraint of Eq. (9c) optimistic: it reserves slightly less battery than the nominal level α requires. The magnitude of this effect is bounded analytically. For a route traversing k arcs with non-negative average pairwise residual correlation $\bar{\rho}$, the true cumulative standard deviation is inflated by at most a factor $\sqrt{1 + (k-1)\bar{\rho}}$ relative to the independence estimate $\sqrt{\sigma^{2,k}_{cum,j}}$. This worst case is attained only under uniform correlation across all arcs of the route; because residual correlation decays with spatial and temporal separation between arcs, the practical inflation is substantially smaller, and conditioning each prediction on speed, temperature, payload, and road grade (as argued above) removes most of the shared-covariate component that would otherwise drive such correlation. When residual correlation is nonetheless suspected — for example under widespread adverse weather — the conservative correction introduced with Eq. (9), which replaces $\sigma^{2,k}_{cum,j}$ by $\sigma^{2,k}_{cum,j} + 2\sum_{p<q} \rho_{pq}\sigma_p\sigma_q$, reinstates the nominal coverage at negligible computational cost whenever the pairwise correlations ρ_{pq} can be estimated. A fully correlated treatment — joint Gaussian processes over the route or copula-based dependence structures — would relax the assumption entirely and is identified as a direction for future work (Section 7).

Chance-Constrained Feasibility Checking (3.6 pp): This is the configuration step between "GAT + uncertainty-aware energy" and the full model (without MILP) in Table 5: activating the chance-constrained feasibility check turns the predicted intervals into proactive charging decisions, and it accounts for the 3.6 pp marginal increment.

MILP Refinement (0.6pp on the medium-scale ablation set): Mathematical optimization addresses limitations of autoregressive neural construction through three mechanisms: (1) vehicle-customer reassignment for improved load balance (67% of instances, average 2.4 reassignments); (2) charging station switching to minimize detours (18% of instances); (3) time window shifting for better charging opportunities (42% of instances). Table 6 shows refinement gap scales with problem complexity.

5.3.2. Component interactions

GAT and uncertainty-aware prediction exhibit additive contributions (6.4 + 3.9 = 10.3 pp combined), indicating orthogonal mechanisms: GAT improves spatial reasoning while uncertainty modeling enhances temporal risk assessment. However, uncertainty-aware prediction shows dependency on GAT: when paired with MLP encoders, uncertainty modeling contributes substantially less than the 3.9 pp it provides with GAT. This arises because effective use of confidence intervals requires spatially adaptive decisions — recognizing when to route toward charging stations based on cumulative variance — which MLP encoders cannot exploit without relational reasoning capacity.

5.4. Model interpretability analysis

Fig. 2 visualizes GAT attention weight distributions under varying battery states, confirming the adaptive behavior identified in the ablation analysis: broad attention across distant customers at high battery (>70%) shifting to concentrated focus on nearby charging stations at low battery (<30%), with dynamic evolution across construction steps from exploration to exploitation.

Energy prediction calibration (Fig. 3) confirms that predicted 95% confidence intervals contain 94.7% of actual observations across 1000 test routes, with heteroscedastic behavior: highway segments exhibit wider intervals (CV = 38%) while urban segments show narrower intervals (CV = 22%). The prediction error distribution (MAPE = 3.8%) is consistent with the approximately Gaussian assumption underlying the chance-constrained formulation. We note, however, that formal distributional diagnostics — quantile–quantile plots, normality tests (e.g., Shapiro–Wilk), and stress tests under heavy-tailed or multimodal deviations — were not performed in the present study; we identify these as a limitation and a priority for future work (Section 7).

5.5. Key findings

Taken together, the computational experiments establish three conclusions that motivate the real-world deployment study. First, integrating calibrated uncertainty quantification with learned routing policies produces consistent improvements over the learning baselines, confirming that neither component alone is sufficient. Second, the performance advantage is substantially larger under dynamic operational conditions than under static benchmarks, suggesting that uncertainty-awareness becomes increasingly valuable as real-world variability increases. Third, the ablation analysis reveals that each architectural component addresses a distinct failure mode, with no single component dominating; this orthogonality suggests the framework is robust rather than dependent on any one design choice.

6. Real-world deployment and sustainability impact

Computational validation establishes technical superiority on benchmark instances drawn from the same distribution as training data. Real-world deployment provides a more demanding test: the framework must transfer effectively to an unseen operational domain with different traffic patterns, charging infrastructure, and energy distributions. We deployed our framework with a mid-sized logistics operator managing 25 electric vehicles (150 kWh, 3-ton capacity) over 13 weeks, serving 60–80 daily customers across 450 km^2 with 12 charging stations. Transfer learning reduced model adaptation from 72 to 4 h using 423 historical routes, confirming that the three-stage protocol successfully bridges the domain gap. Integration with existing transportation management systems proceeded through four phases: infrastructure setup (6 weeks), system training (4 weeks), pilot expansion (8 vehicles, 4 weeks), and full deployment (25 vehicles, 4 weeks).

6.1. Operational performance

Table 7 demonstrates sustained performance improvements validating computational findings in operational practice. The *Baseline* column reflects the operator's pre-deployment routing operations, which relied on a commercial legacy transportation management system using rule-based manual planning without mathematical optimization or machine learning components. Daily costs decreased 16.9% relative to the legacy rule-based baseline (approximately €143,000 in projected



Fig. 2. GAT attention weight distributions during routing decisions. (a) High battery state (>70%): attention distributed across distant customers. (b) Low battery state (<30%): attention concentrated on nearby charging stations. (c) Attention evolution across eight construction steps.

Table 7
Deployment performance evolution across 13 weeks. Metrics averaged over 5-day operational weeks.

Metric	Baseline	Week 1–4	Week 5–8	Week 9–13	Improvement
Daily cost (€)	2847	2643	2489	2367	−16.9%
On-time delivery	91.3%	94.2%	96.5%	98.1%	+6.8 pp
Energy (kWh)	1456	1342	1294	1231	−15.5%
Planning time	45 min	2.3 min	2.1 min	1.8 min	−96.0%

annual savings, extrapolated from the 13-week pilot assuming seasonal stationarity; 4.2-month payback). This operational saving reflects the combined value of algorithmic improvement and system modernization; the framework's algorithmic advantage is separately established by the 8.8% cost reduction over the strongest learning baseline (MV-MoE) in Section 5. On-time delivery improved to 98.1% (+6.8 pp), energy consumption declined 15.5% (182 kWh/day), and planning time collapsed 96% (45 min to 1.8 min). Progressive week-over-week improvement reflects online learning as policies adapted to local traffic patterns, driver behaviors, and seasonal conditions.

Note that on-time delivery (Table 7) captures the proportion of customers served within time windows during operations, whereas feasibility (Table 3) measures the percentage of routes satisfying battery constraints at the 95% confidence level. The higher deployment metric reflects transfer learning adaptation and online policy refinement over 13 weeks.

6.2. Limitations observed in practice

Alongside the performance improvements documented above, the 13-week deployment revealed practical limitations that are important to report.

MILP refinement reliability. The optional MILP refinement phase (Section 4.4) frequently timed out without improving upon the neural solution, particularly for large instances (80–100 customers) where the branch-and-cut search space is extensive. In these cases, the framework correctly fell back to the DRL-Pure solution, preserving real-time operational capability. However, this behavior means the 2.8% average improvement attributed to MILP refinement across the 30 benchmark instances (Table 3), rising to 4.1% on the largest instances (80–100 customers; Table 6), should be interpreted as an average across instances where refinement succeeds, not a guarantee across all problem sizes. Practitioners should treat the MILP phase as a best-effort offline enhancement rather than a reliable quality guarantee.

Early deployment instability. Table 7 reveals that performance improvement was progressive rather than immediate: costs in Weeks 1–4 (€2643/day) remained meaningfully above the final steady-state level (€2367/day, Weeks 9–13), reflecting the time required for online

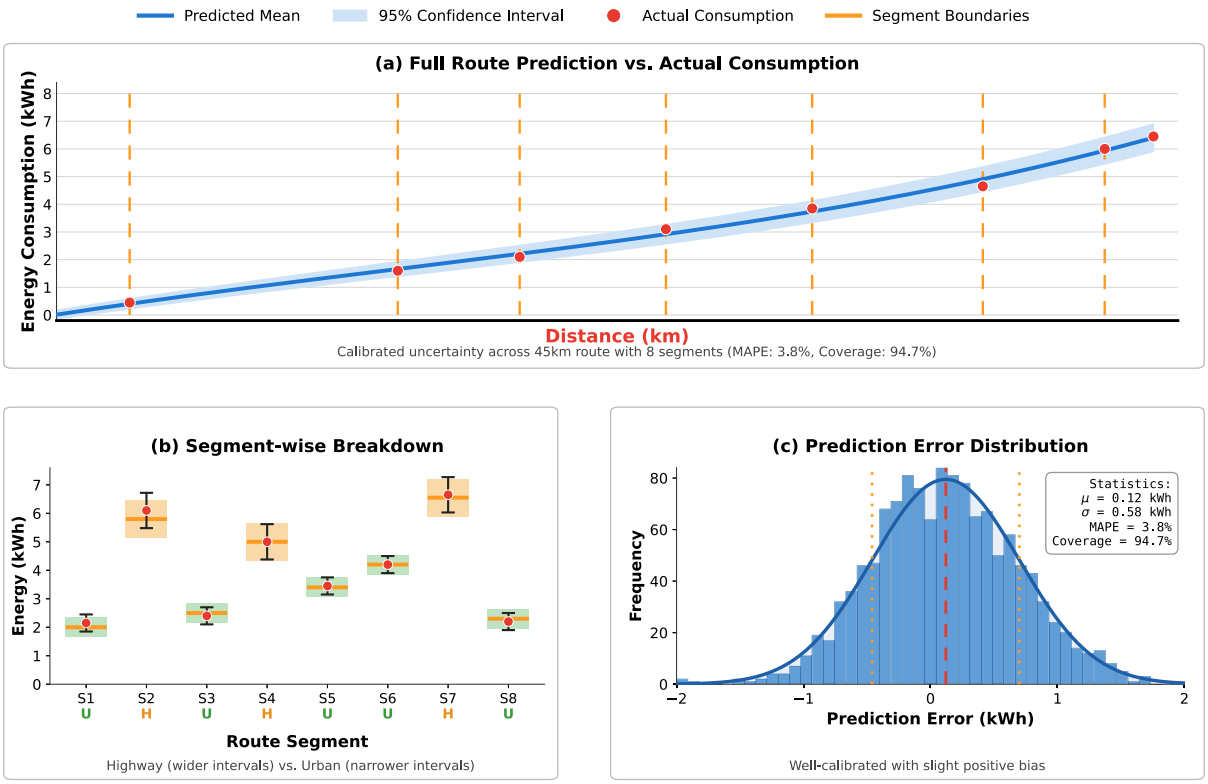


Fig. 3. Energy consumption prediction with 95% confidence intervals. (a) Full route prediction versus actual consumption (94.7% empirical coverage). (b) Segment-wise breakdown: highway versus urban segments. (c) Prediction error distribution (MAPE = 3.8%).

learning to adapt to local traffic patterns and driver behaviors. Operators considering deployment should anticipate a stabilization period of approximately four weeks before full performance benefits are realized.

System resilience validated practical viability under operational disruptions: highway closure (8.4 s replanning, 84% faster than baseline), charging station outage (zero strandings versus 3 historical incidents), and +28% demand surge (absorbed in 4.2-minute reoptimization maintaining 96.3% on-time delivery). Driver satisfaction surveys (Likert scale, $n = 8$; interpreted as descriptive evidence given the small sample) indicated notable improvements: route quality (3.2 ± 0.8 to 4.6 ± 0.3), range prediction reliability (2.8 ± 0.9 to 4.7 ± 0.4), and work-life balance (2.4 ± 0.9 to 4.1 ± 0.6).

6.3. Multi-dimensional sustainability

Environmental impact. Carbon emissions analysis comparing the diesel baseline (128.5 tons CO_2/year) with the electric DRL-Hybrid fleet (133.9 tons) reveals approximate carbon parity on Italy's transitional grid (0.418 kg CO_2/kWh); relative to the strongest learning baseline, uncertainty-aware routing reduces energy use by 3.5% (Section 5), with a commensurate reduction in operational emissions. While this parity limits immediate operational carbon savings, three factors establish strategic value: (1) elimination of local $PM_{2.5}$ (0.73 tons/year) and NO_x (2.14 tons/year) in delivery zones, valued at €47,300 in avoided healthcare costs [55]; (2) automatic emissions reduction to 87.3 tons by 2030 as grid decarbonizes to 0.273 kg CO_2/kWh without fleet modifications; (3) noise pollution reduction (58 dB vs. 73 dB diesel) enabling extended delivery windows (6:00–22:00). Energy efficiency improved continuously: consumption declined from 1.89 to 1.52 kWh/package (19.6%), while strategic charging shifted 67% of sessions to off-peak hours (vs. 51% baseline), reducing grid stress.

Economic viability. Operator savings (approximately €143,000 annually) represent the primary, directly accountable economic benefit; additional externalities such as local air-quality improvements and grid-optimization effects provide further societal value that is not captured in operator accounts. Over a five-year horizon, total cost of ownership is favorable relative to diesel, driven primarily by lower energy and maintenance costs; per-vehicle economics are expected to improve with fleet scale and shared charging infrastructure through standard economies of scale, although the precise magnitude depends on local vehicle, energy, and infrastructure prices and is not quantified from the single-operator pilot.

Social dimensions and scalability. Dispatcher workflow transformation freed 6.5 h weekly per dispatcher. Simulation projects that planning time scales approximately linearly with fleet size (0.8 s to 13.4 s, $R^2 = 0.97$). Transfer learning across four regions achieves 94% of full-training performance with 93% time reduction.

7. Discussion and conclusions

This work demonstrates that integrating uncertainty-aware deep reinforcement learning with mathematical optimization can help address the tension between environmental mandates for rapid decarbonization and operational complexities that have constrained electric vehicle adoption in urban logistics. The convergence of computational performance (8.8% cost reduction over the strongest learning baseline, 12.2% improvement under dynamic conditions), operational validation (approximately €143,000 in projected annual savings, 98.1% on-time delivery, 96% planning time reduction), and multi-dimensional sustainability impact (approximate carbon parity with diesel on the current grid alongside elimination of local air pollutants, improved driver satisfaction) provides evidence that several technological barriers to sustainable urban logistics can be addressed through uncertainty-aware optimization, while the broader transition also depends on infrastructure, policy, and market factors that lie beyond the scope of routing. The hybrid paradigm — combining neural networks' pattern recognition with mathematical optimization's solution quality guarantees

— offers a template for problems requiring both rapid inference and high-quality solutions across diverse operational contexts.

The 13-week operational study indicates that a meaningful share of the societal value — local air-quality and grid externalities — falls outside operator accounts and is not internalized by current market structures. This finding has direct policy implications: carbon pricing mechanisms, performance-based incentives for zero-emission delivery, and coordinated public charging infrastructure could accelerate urban fleet electrification from current 8%–12% levels to 40%–60% within 5–7 years. The transfer learning protocol achieving 94% of full-training performance with 93% time reduction demonstrates that the model-adaptation component of deployment effort can be substantially lowered (cf. Section 6), lowering one barrier to adoption for operators who lack machine learning expertise.

This study has limitations that suggest opportunities for extension. The driver satisfaction component relied on a small convenience sample ($n = 8$), limiting statistical inference; findings should be interpreted as qualitative indicators. The framework assumes homogeneous fleets and next-day planning horizons; heterogeneous operations and same-day delivery require multi-agent architectures and streaming algorithms. Environmental assessment focuses on operational emissions, excluding manufacturing and battery lifecycle impacts. The 13-week deployment in a single region limits generalizability across different grid compositions and seasonal patterns. In addition, the chance-constrained feasibility guarantee is conditional on the Gaussian arc-level energy model and on the calibration of the learned uncertainty estimator; the effect of correlated residuals is analyzed in Section 5.3, whereas rigorous distributional validation of the Gaussian assumption — quantile–quantile plots, normality tests, and heavy-tailed stress tests — was not performed in the present study and is identified as a priority for future work, alongside the conservative correction and the joint-uncertainty models noted below. Future research could address these limitations through multi-agent extensions for heterogeneous fleets, streaming architectures for same-day dynamic requests, explicit modeling of charger congestion and queueing at shared stations, hierarchical reinforcement learning for infrastructure-routing co-optimization, joint correlated-uncertainty modeling (route-level Gaussian processes or copula-based dependence structures, complemented by Bonferroni-style aggregation of the individual chance constraints into a conservative joint route-level guarantee), and comprehensive lifecycle frameworks integrating manufacturing, degradation, and circular economy considerations.

CRediT authorship contribution statement

Aymen Jalil Abdulelah: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Emrullah Sonuç:** Writing – review & editing, Validation, Supervision, Investigation, Conceptualization. **Esam Taha Yassen:** Writing – review & editing, Validation, Supervision, Investigation, Conceptualization. **Mohammed Nasser Al-Andoli:** Writing – review & editing, Resources, Formal analysis, Data curation.

Code availability

The full source code for this work — including the GAT encoder, LSTM-based probabilistic energy predictor, PPO policy, chance-constraint checker, MILP solver wrapper, training and evaluation pipelines, transfer learning protocol, and reproducible result generation scripts — is publicly available under the MIT License at:

https://github.com/Aymen-Anbar/evrptw-repo

The repository includes pre-trained model configuration details, all 16 result figures, 17 data tables (CSV and Excel), and a machine-readable JSON archive of every quantitative result reported in this paper.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The benchmark instances used in this study are derived from the publicly available Solomon VRPTW benchmarks [53], following the EVRPTW extension of Schneider et al. [11]. Pre-generated experimental results, data tables, and figures are available in the code repository described in the Code Availability section below. The real-world deployment dataset is not publicly available due to confidentiality obligations with the logistics operator.

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